Boundary Laws for Luce Rankings

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Mechanical extraction — proof audit pending

Abstract

A Luce ranking is generated by ordering independent exponential clocks with prescribed positive rates. This paper gives exact finite identities for the top of the ranking and sharp limiting laws for the bottom. The top-window law has an exact total-variation formula equal to the collision probability of an associated product sample, together with an exact one-sided relative defect. For countably many clocks, the paper identifies the coordinatewise limiting bottom-window subprobability law and computes its missing mass exactly as a terminal leakage event. A vanishing-block construction also shows why permuton convergence alone cannot control microscopic cycle statistics.

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1 Status and provenance

This source is a mechanical extraction from the frozen 139-page Luce manuscript at repository commit `6178a99a8f2ee7b6416edd026e7780b9b8fa9fa0`. The mathematical blocks between the archival sentinels below are copied byte-for-byte from the frozen source. Only the standalone preamble, abstract, section wrappers, provenance notice, and any cited bibliography entries have been added.

The extraction preserves every theorem hypothesis, identity, limiting statement, quantifier, and proof step in the selected boundary package. It does not audit, repair, strengthen, weaken, or modernize the mathematics. Its status remains a working mechanical extraction until a separate adversarial audit is completed.

2 Exponential-race input

2.1 The exponential race

Let $\theta_1, \dots, \theta_n > 0$, and let $X_i \sim \mathrm{Exp}(\theta_i)$ be independent. For distinct labels $i_1, \dots, i_n$,

$$\mathbb{P}(X_{i_1} < \cdots < X_{i_n}) = \prod_{r=1}^{n} \frac{\theta_{i_r}}{\theta_{i_r} + \theta_{i_{r+1}} + \cdots + \theta_{i_n}}.$$

This is the standard exponential race identity and follows from the memoryless property.

3 A microscopic obstruction

Proposition 3.1 (Vanishing-block implantation)**.** *Let $m_n \to \infty$ with $m_n = o(n)$, and let $\pi_n \in S_{m_n}$ be arbitrary. There are Luce weights $\theta_{n,1}, \dots, \theta_{n,n}$ such that*

$$\mathbb{P}\big(\sigma_n(i) = \pi_n(i) \text{ for every } 1 \le i \le m_n\big) \longrightarrow 1,$$

while the step profiles

$$f_n(x) = \theta_{n,\lceil nx \rceil}, \qquad 0 < x \le 1,$$

converge almost everywhere to the constant profile $f \equiv 1$. Consequently, σ_n has the uniform permuton limit, but its cycle structure on $m_n = o(n)$ prescribed labels can be chosen arbitrarily.

Proof. Write $\tau_r = \pi_n^{-1}(r)$, $1 \le r \le m_n$, and choose R_n so that $R_n/n \to \infty$. Set

$$\theta_{n,\tau_r} = R_n^{m_n - r + 1}, \qquad 1 \le r \le m_n, \qquad \theta_{n,i} = 1, \quad i > m_n.$$

Let $X_{n,i} \sim \mathrm{Exp}(\theta_{n,i})$ be independent, and let $Y_n = \min_{i > m_n} X_{n,i} \sim \mathrm{Exp}(n - m_n)$. For $1 \le r < m_n$,

$$\mathbb{P}(X_{n,\tau_r} \ge X_{n,\tau_{r+1}}) = \frac{1}{R_n + 1},$$

while

$$\mathbb{P}(X_{n,\tau_{m_n}} \ge Y_n) = \frac{n - m_n}{R_n + n - m_n}.$$

A union bound therefore gives

$$\begin{aligned} &\mathbb{P}\big(X_{n,\tau_1} < \cdots < X_{n,\tau_{m_n}} < Y_n\big) \\ &\qquad \ge 1 - \frac{m_n - 1}{R_n + 1} - \frac{n - m_n}{R_n + n - m_n} \longrightarrow 1, \end{aligned}$$

because $n/R_n \to 0$ and $m_n/R_n = (m_n/n)(n/R_n) \to 0$. On this event, the first m_n labels in the Luce draw order are $\tau_1, \dots, \tau_{m_n}$, so $\sigma_n(\tau_r) = r = \pi_n(\tau_r)$ for every r.

For every fixed $x > 0$, the index $\lceil nx \rceil$ exceeds m_n for all large n, and hence $f_n(x) = 1$. Thus $f_n \to 1$ almost everywhere. On the high-probability event above, the remaining $n - m_n$ labels all have equal weight and therefore appear in a uniformly random order after the implanted block. The implanted points carry permuton mass $m_n/n = o(1)$, while a uniform permutation of the remaining labels has the uniform permuton limit. Hence the whole permutation has the uniform permuton limit. The final assertion follows by prescribing the cycle decomposition of π_n. $\square$

4 Top windows: exact boundary identities

Normalize $\sum_i \theta_i = 1$. Let P_k be the law of the first k labels under Luce sampling without replacement, and let Q_k be the product law of k independent draws from $[n]$ with probabilities θ_i. Define

$$\mathcal{D}_k = \{(i_1, \dots, i_k) \in [n]^k : i_1, \dots, i_k \text{ are pairwise distinct}\}$$

and

$$e_k(\theta_1, \dots, \theta_n) = \sum_{1 \le j_1 < \cdots < j_k \le n} \theta_{j_1} \cdots \theta_{j_k}.$$

Theorem 4.1 (Exact top-window total variation)**.** *For every $k \le n$,*

$$\|P_k - Q_k\|_{\mathrm{TV}} = Q_k(\mathcal{D}_k^c) = 1 - k! e_k(\theta_1, \dots, \theta_n).$$

Proof. The measure P_k is supported on $\mathcal{D}_k$. If $i = (i_1, \dots, i_k) \in \mathcal{D}_k$, then

$$P_k(i) = \prod_{r=1}^{k} \frac{\theta_{i_r}}{1 - \theta_{i_1} - \cdots - \theta_{i_{r-1}}},$$

with the empty sum equal to zero. Hence $P_k(i) \ge Q_k(i)$ on $\mathcal{D}_k$, while $P_k(i) = 0$ on $\mathcal{D}_k^c$. The positive and negative parts therefore have equal mass, and

$$\begin{aligned} 2\|P_k - Q_k\|_{\mathrm{TV}} &= \sum_{i \in \mathcal{D}_k} (P_k(i) - Q_k(i)) + \sum_{i \notin \mathcal{D}_k} Q_k(i) \\ &= 2Q_k(\mathcal{D}_k^c). \end{aligned}$$

Thus $\|P_k - Q_k\|_{\mathrm{TV}} = 1 - Q_k(\mathcal{D}_k)$. Finally,

$$Q_k(\mathcal{D}_k) = \sum_{\{j_1, \dots, j_k\}} k! \theta_{j_1} \cdots \theta_{j_k} = k! e_k(\theta_1, \dots, \theta_n).$$

□

Define the one-sided relative defect

$$d_\infty^+(P_k, Q_k) = \left\| 1 - \frac{d(Q_k|_{\mathcal{D}_k})}{dP_k} \right\|_{L^\infty(P_k)} = \max_{i \in \mathcal{D}_k} \left(1 - \frac{Q_k(i)}{P_k(i)} \right).$$

This is not a metric. Let $\theta_{(1)} \ge \cdots \ge \theta_{(n)}$ be the decreasing rearrangement.

Theorem 4.2 (Exact one-sided relative L^∞ defect)**.** *For $k \le n$,*

$$d_\infty^+(P_k, Q_k) = 1 - \prod_{r=1}^{k-1} \left(1 - \sum_{j=1}^{r} \theta_{(j)} \right),$$

where the empty product is one when $k = 1$.

Proof. For $i = (i_1, \dots, i_k) \in \mathcal{D}_k$,

$$\frac{Q_k(i)}{P_k(i)} = \prod_{r=1}^{k-1} \left(1 - \sum_{j=1}^{r} \theta_{i_j}\right).$$

For every r, $\sum_{j=1}^{r} \theta_{i_j} \le \sum_{j=1}^{r} \theta_{(j)}$. Since each factor $1 - x$ is decreasing in x, the product is minimized by taking the first $k-1$ selected weights to be $\theta_{(1)}, \dots, \theta_{(k-1)}$. The final selected label does not enter the product. This proves the formula. □

Corollary 4.3 (Birthday transfer)**.** *Let $\xi_1, \dots, \xi_k$ be independent with law θ, and let*

$$W_k = \sum_{1 \le r < s \le k} \mathbf{1}_{\{\xi_r = \xi_s\}}.$$

Then

$$\|P_k - Q_k\|_{\mathrm{TV}} = \mathbb{P}(W_k \ge 1).$$

Consequently, every asymptotic estimate for the no-collision probability $\mathbb{P}(W_k = 0)$, equivalently for the collision probability $\mathbb{P}(W_k \ge 1)$, transfers immediately to the top-window total-variation distance.

Proof. Under Q_k, the event $\{W_k = 0\}$ is precisely $\mathcal{D}_k$. Hence

$$Q_k(W_k = 0) = Q_k(\mathcal{D}_k) = k! e_k(\theta_1, \dots, \theta_n).$$

Theorem 4.1 therefore gives

$$\|P_k - Q_k\|_{\mathrm{TV}} = 1 - Q_k(\mathcal{D}_k) = Q_k(W_k \ge 1),$$

which is the claimed identity. □

5 Bottom windows and terminal leakage

Let $(\theta_i)_{i \ge 1}$ be positive rates and let $X_i \sim \mathrm{Exp}(\theta_i)$ be independent. Define

$$f(x) = \sum_{i=1}^{\infty} e^{-\theta_i x}, \qquad x_0 = \inf\{x : f(x) < \infty\}.$$

The function f is nonincreasing. For a finite set $A \subset \mathbb{N}$ and $x > 0$, define

$$\prod_{i \notin A} (1 - e^{-\theta_i x}) := \lim_{N \to \infty} \prod_{\substack{i \le N \\ i \notin A}} (1 - e^{-\theta_i x}).$$

The limit always exists. It is positive exactly when $\sum_{i \notin A} e^{-\theta_i x} < \infty$, and otherwise it equals zero. Indeed, for $u_i = e^{-\theta_i x} \in (0, 1)$, the partial products are decreasing, and $\log(1 - u_i) \sim -u_i$ once $u_i \to 0$; if u_i does not tend to zero, the product plainly vanishes.

For distinct $a_1, \dots, a_k$, define

$$\mu_k(a_1, \dots, a_k) := \int_{x_1 > \cdots > x_k > 0} \prod_{j=1}^{k} \theta_{a_j} e^{-\theta_{a_j} x_j} \prod_{i \notin \{a_1, \dots, a_k\}} (1 - e^{-\theta_i x_k})\, dx_1 \cdots dx_k.$$

The following finite-volume statement identifies μ_k as an actual limiting subprobability law rather than only a formal candidate.

Proposition 5.1 (Finite bottom-window convergence)**.** *For $N \geq \max_j a_j$, let $\mu_k^{(N)}(a_1, \ldots, a_k)$ be the probability that the k largest values among $X_1, \ldots, X_N$, listed in decreasing order, have labels $a_1, \ldots, a_k$. Then, for every fixed ordered distinct tuple,*

$$\mu_k^{(N)}(a_1, \ldots, a_k) \longrightarrow \mu_k(a_1, \ldots, a_k).$$

Thus μ_k is the coordinatewise limit of the finite bottom-window laws.

Proof. For $N \geq \max_j a_j$,

$$\mu_k^{(N)}(a_1, \ldots, a_k) = \int_{x_1 > \cdots > x_k > 0} \prod_{j=1}^{k} \theta_{a_j} e^{-\theta_{a_j} x_j} \times \prod_{\substack{1 \leq i \leq N \\ i \notin \{a_1, \ldots, a_k\}}} (1 - e^{-\theta_i x_k})\, dx_1 \cdots dx_k.$$

The finite products decrease pointwise to the defining infinite product and are bounded by one. The remaining factor is the joint density of the specified independent clocks and is integrable over the chamber. Dominated convergence proves the assertion. □

Theorem 5.2 (Terminal leakage)**.** *Assume $x_0 < \infty$. Let*

$$N_0 = \sum_{i=1}^{\infty} \mathbf{1}_{\{X_i > x_0\}}.$$

Then

$$\sum_{a_1, \ldots, a_k}^{*} \mu_k(a_1, \ldots, a_k) = \mathbb{P}(N_0 \geq k),$$

where $\sum^*$ denotes summation over distinct ordered k-tuples. In particular, if $f(x_0) = \infty$, then the mass is one. If $f(x_0) < \infty$, the escaped mass is $\mathbb{P}(N_0 < k) > 0$, so the limiting law is strictly defective.

Proof. For every $x > x_0$, $f(x) < \infty$. Hence

$$\sum_i \mathbb{P}(X_i > x) = \sum_i e^{-\theta_i x} < \infty,$$

so Borel–Cantelli implies that only finitely many of the events $\{X_i > x\}$ occur almost surely. Applying this with $x = x_0 + 1/m$, $m \geq 1$, shows that the point set $\{X_i : X_i > x_0\}$ is locally finite in (x_0, ∞). Thus its largest, second largest, and subsequent points are well-defined whenever $N_0 \geq k$; possible accumulation can occur only at the endpoint x_0, not above it. If $x_0 = 0$, then $N_0 = \infty$ almost surely, since $X_i > 0$ almost surely for every i, and the same argument applies with boundary $x_k = 0$.

For a fixed ordered distinct k-tuple $a = (a_1, \ldots, a_k)$, the event that the largest k clock values above x_0, equivalently the ordered bottom-window labels, are $a_1, \ldots, a_k$ and have heights in $dx_1 \cdots dx_k$, with $x_1 > \cdots > x_k > x_0$, has density

$$\prod_{j=1}^{k} \theta_{a_j} e^{-\theta_{a_j} x_j} \prod_{i \notin \{a_1, \ldots, a_k\}} (1 - e^{-\theta_i x_k}).$$

The first factor fixes the specified heights and the product says that every other variable is below x_k. Conversely, if this density contributes with $x_k > x_0$, then these are exactly the largest k clock values above x_0. If $x_k < x_0$, the infinite product is zero because $f(x_k) = \infty$. The hyperplane $x_k = x_0$ has Lebesgue measure zero. Therefore integrating and summing over all ordered distinct a's partitions the event $\{N_0 \geq k\}$, giving the identity.

If $f(x_0) = \infty$, then $\sum_i \mathbb{P}(X_i > x_0) = \infty$, so independence and the second Borel–Cantelli lemma imply $N_0 = \infty$ almost surely. If $f(x_0) < \infty$, then necessarily $x_0 > 0$, and N_0 is a finite Poisson-binomial variable with parameters $e^{-\theta_i x_0}$. Since these parameters are all strictly less than one and have finite sum,

$$\mathbb{P}(N_0 = 0) = \prod_{i=1}^{\infty}(1 - e^{-\theta_i x_0}) > 0.$$

Hence $\mathbb{P}(N_0 < k) > 0$. □

Remark 5.3 (Complete leakage when $x_0 = \infty$). If $x_0 = \infty$, then $f(x) = \infty$ for every finite x. The infinite product in the definition of μ_k therefore vanishes for every point of the integration chamber, so $\mu_k \equiv 0$. In this case all finite-label coordinate mass escapes.

Corollary 5.4 (A log-log phase diagram)**.** *Suppose, as $i \to \infty$,*

$$\theta_i = \beta \log i + \alpha \log\log i + O(1), \qquad \beta > 0.$$

Then $x_0 = 1/\beta$, and the bottom fixed-window law is non-defective if and only if $\alpha \leq \beta$.

Proof. For large i,

$$e^{-\theta_i x} \asymp i^{-\beta x}(\log i)^{-\alpha x}.$$

Thus $\sum_i e^{-\theta_i x}$ converges for $x > 1/\beta$ and diverges for $x < 1/\beta$, so $x_0 = 1/\beta$. At x_0,

$$f(x_0) \asymp \sum_i \frac{1}{i(\log i)^{\alpha/\beta}},$$

which diverges exactly when $\alpha/\beta \leq 1$. Apply Theorem 5.2. □